Seminar Report on Express: Lowering the Cost of Metadata-hiding Communication with Cryptographic Privacy

A. Student (*remember to remove ID for the first draft and peer review!*)

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Abstract

This report studies the reproducibility and performance of Express, a system for metadata-hiding communication. Using two servers, express provides cryptographic security against at most one malicious server and an arbitrary number of malicious clients. The paper presents new protocols that offer low communication overhead, regardless of the number of users, and improved computation costs using symmetric key cryptography. Compared to Pung and Riposte, two previous cryptographically secure systems, express lowers cost and increases message throughput. This makes Express an ideal applications for communication between whisteblowers and journalists.

1 Introduction

In todays digital landscape, secure communication has become increasingly crucial, with growing capabilites of surveillance and metadata analysis. Secure messaging apps like Signal and Transport Layer Security (TLS) encrypt the content of sent messages, but they do not protect metadata. It is possible to observe who sent and received messages, when and how often they communicated and how much data they exchanged. These information are as sensitive as the message itself and should therefore not be leaked. An option for metadata hiding would be using a TOR browser or SecureDrop, but they are vulnerable to attacks by adversaries that control many network points. Other existing systems to hide metadata either have to high communication cost to be used in reality, or they are have low computation cost but their security gets worse with every round of communication, also making them impractical.
While two servers are required, express still runs with one malicious server. Express' target audience are journalists and whistleblowers. To receive messages, a journalist, also client in the client-server architecture, registers a mailbox with the two servers.

2 Background

Theorem 1 (Incompleteness theorem, Gödel, 1931)**.**
If c be a given recursive, consistent class of formulae, then the propositional formula which states that c is consistent is not c-provable; in particular, the consistency of P is unprovable in P, it being assumed that P is consistent.

2.1 First background topic

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa. (See Figure 1.)

Figure 1: An example privacy mechanism

Header line	Column 2	Column 3
Content line	2	3
Another content line to showcase wrapping of table cells	2	3

Table 1: An example table

2.2 Second background topic

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3 Main content

Note that you *must* rename this section to indicate *what the content is*.

4 Another topic in the main content

Your “main content” can be split into as many sections as needed.

5 Related work

Pung: Pung also focuses on metadata hiding in communication systems.

Here, you include short descriptions of similar work and explain *how* they are different from the main content you have presented.

Level of detail similar to an abstract.

Note that you may include the related work as a subsection in the Background Section 2.

6 Conclusion

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

7 Use of generative digital tools

The use of AI tools must always be declared completely and transparently. For examples of how to do this, please see the DMI guidance: `https://dmi.unibas.ch/fileadmin/user_upload/dmi/Studium/Computer_Science/Diverses/Wegleitungen/Dokumente_2023/Guidance_on_AI_in_teaching_in_Computer_Science.pdf`.

References

Gödel, K. (1931). Über formal unentscheidbare sätze der principia mathematica und verwandter systeme i. *Monatshefte für mathematik und physik*, 38(1):173–198.

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September 2023